

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5301
Subject Title	Service Design and Fleet Management for Low-altitude Economy
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject will provide students with 1. Advanced knowledge in facility location, service network design, fleet assignment, flight scheduling, and vehicle routing of low-altitude transport; 2. Comprehensive understanding of multi-modal transportation within low-altitude economy; and 3. Operational planning skills for unmanned aerial vehicles in various applications.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Design and develop mathematical models to solve problems related to facility location, service network design, fleet assignment, flight scheduling, and vehicle routing in low-altitude transport; b. Apply various strategies and techniques to analyse and optimize multi-modal transportation in a low-altitude economy; and c. Utilize decision support methods to enhance the efficiency of unmanned aerial vehicle applications.
Subject Synopsis/ Indicative Syllabus	Operations management in low-altitude transport: Decisions on helipad and vertiport locations, construction of service networks, fleet assignments, flight timetable design, queueing analysis, and vehicle routing for drones, electric vertical take-off and landing (eVTOL) aircraft, and other aerial vehicles. Multi-modal transportation: Game theory analysis, analysis of collaboration and competition between different transport modes, synchronization of timetables, and strategies for demand analysis and pricing in low-altitude transport. Service design in special context: Complex network theory, point-to-point/hub-and-spoke, network model; helipad/miniport for eVTOL; Derivation of objectives and constraints for drone applications in parcel delivery and rescue logistics, model construction, and implementation.

	Moreover, the background knowledge of the business planning and project management will also be introduced.				
Teaching/Learning Methodology	Teaching is conducted through lectures and assignments. The basic knowledge, research methodology and theoretical models will be introduced. The understanding of how to address and formulate problems by using mathematical programming, data analytics, and operations research techniques is emphasised.				
	Teaching/Learning Methodology	Outcomes			
		a	b	c	
	Lecture	√	√	√	
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)		
			a	b	c
	1. Assignment	30%	√	√	√
	2. Course Project	20%			√
	3. Final examination	50%	√	√	√
	Total	100%			
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.5 × Continuous Assessment + 0.5 × Final Examination The continuous assessment (50%) is aimed at enhancing the students’ comprehension and assimilation of various topics of the syllabus via assignments and a course project. The final examination (50%) will also be considered to assess the students' learning outcomes.				
Student Study Effort Expected	Class contact:				
	▪ Lecture/Case Study			39 Hrs.	
	Other student study effort:				
	▪ Self-learning/preparation			66 Hrs.	
	Total student study effort			105 Hrs.	
Reading List and References	1. Hillier F S, Lieberman G J. Introduction to operations research: 11th Edition [M]. McGraw-Hill, 2021. 2. Wandelt S, Wang S, Zheng C, et al. AERIAL: A Meta Review and Discussion of Challenges Toward Unmanned Aerial Vehicle				

	Operations in Logistics, Mobility, and Monitoring[J]. IEEE Transactions on Intelligent Transportation Systems, 2023. 3. Bazargan M. Airline operations and scheduling[M]. Routledge, 2016. 4. Boyd S P, Vandenberghe L. Convex optimization[M]. Cambridge University Press, 2004. 5. Button K. Transport economics: 4th Edition [M]. Edward Elgar Publishing, 2022. 6. Hart W E, Laird C D, Watson J P, et al. Pyomo — Optimization Modeling in Python: 3rd Edition [M]. Berlin: Springer, 2021.
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